Stats for Sociologists

Course: SOC6302H1F

Instructor: Professor Pepin

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INSTRUCTIONS:

Fill out the first page of the scantron. Confirm you have included:

- Your student number (write the number and fill in the ovals)
- Your surname and first initial (write the letters and fill in the ovals)

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1. Using 2024 GSS data, a researcher estimated the hours per day Americans spend relaxing. Based on the 95% confidence interval from R, which interpretation is correct?

sex	n	mean	sd	lowCI	hiCI
male	1139	4.2	3.7	4.0	4.5
female	1174	4.0	3.5	3.8	4.2

   (a) We cannot determine whether the null hypothesis of zero difference between the parameters will be rejected at the 0.05 level.
   (b) The confidence intervals overlap, so we can't know for sure if there is a statistical difference between the average daily time men and women spend in leisure.
   (c) We are 95% certain that women have less time per day than men for leisure activities, likely because of their greater time spent on housework and childcare.
   (d) The confidence intervals do not overlap so we can conclude that the average time men and women spend on leisure activities per day are statistically significantly different.

2. Has agreement with the statement that 'Black Americans face worse jobs, income, and housing than white Americans due to discrimination' shifted? Using a significance level of $\alpha = 0.01$, what conclusion should be drawn?

	year		
	2022	2024	Total
opinion			
no	509 (45%)	522 (51%)	1,031 (48%)
yes	619 (55%)	494 (49%)	1,113 (52%)
Total	1,128 (100%)	1,016 (100%)	2,144 (100%)

Pearson's Chi-squared test, p=0.004

   (a) The drop in agreement with the statement between 2022 and 2024 is statistically significant.
   (b) The small change is statistically significant, but most people still agree with the statement.
   (c) The increase in agreement with the statement between 2022 and 2024 is statistically significant.
   (d) There is a statistically significant difference, so we fail to reject the null hypothesis.

3. In *Naked Statistics: Stripping the Dread from the Data?*, Wheelan explains that "most sample means will lie reasonably close to the population mean; the standard error is what defines 'reasonably close.' " This statement implies the standard error:

   (a) measures how much sample means tend to vary around the population mean.
   (b) measures the spread of individual observations around the sample mean.
   (c) increases as sample size increases.
   (d) measures the distance between a particular sample mean and the population mean.

4. National survey data shows that 69% of Canadians support mandatory childhood vaccinations. A local researcher believes that support is likely much higher in Toronto compared with the national average. Which of the following equations represents the research hypothesis?

   (a) $\mu < 0.69$
   (b) $\mu = 0.69$
   (c) $\mu \neq 0.69$
   (d) $\mu > 0.69$

5. In a normally distributed set of test scores, a student scoring below the average will have a _____.

   (a) positive z-score
   (b) positive standard error
   (c) negative standard error
   (d) negative z-score

6. Which RStudio pane is used to write and edit R scripts and Quarto documents?

   (a) Environment
   (b) Files
   (c) Source
   (d) Console

7. What is the purpose of `drop_na()` in dplyr?

   (a) It replaces missing values with zeros.
   (b) It drops columns that are entirely NA.
   (c) It converts NA values to NULL.
   (d) It removes rows that contain missing values.

8. According to Wheelan, in *Naked Statistics: Stripping the Dread from the Data?*, what is the most common cause of inaccurate polling results?

   (a) Misuse of statistical software during data analysis
   (b) Random variation in respondent answers
   (c) Incorrect calculations of standard errors
   (d) Biased sampling methods or poorly worded questions

9. Researchers found that couples report about 10 arguments per month on average. If a couple received a z-score of -2 for this measure, which of the following best describes the amount of arguments for this couple compared with the average?

   (a) This couple had fewer arguments than about 98% of couples.
   (b) This couple had about 8 arguments per month.
   (c) This couple had fewer arguments than about 5% of couples.
   (d) This couple had fewer arguments than about 2% of couples.

10. What does the expression `5 != 3` evaluate to in R?

    (a) NA
    (b) NULL
    (c) FALSE
    (d) TRUE

11. What does the `::` operator do in R?

    (a) Imports all functions from a package
    (b) Specifies a function from a particular package
    (c) Assigns a value to a variable
    (d) Comments out a line of code

12. The level of measurement for the `hapmar` variable is _____

    [1] "Happiness of marriage"

	Freq	%	% Cum.
very happy	931	56.60	56.60
pretty happy	646	39.27	95.87
not too happy	68	4.13	100.00
Total	1645	100.00	100.00

    (a) ordinal
    (b) nominal
    (c) interval
    (d) dichotomous

13. When examining a frequency distribution table, which column most directly identifies the category containing the 50th percentile (median)?

    (a) frequencies
    (b) cumulative percentages
    (c) percentages
    (d) relative frequencies

14. Which description best describes a correlation coefficient of r = -0.62?

    (a) A statistically insignificant relationship
    (b) Weakly negative relationship
    (c) Strong negative relationship
    (d) No relationship exists between the variables

15. Which of the following is a valid use of `group_by()` and `summarize()`?

    (a) data |> summarize(mean = group_by(sex))
    (b) summarize(group_by(data, age), mean(sex))
    (c) data |> group_by(sex) |> summarize(freq = n())
    (d) group_by(data, summarize(sex))

16. In the 1970s, Americans watched an average of 3.05 TV hours per day, compared with an average of 3.3 hours per day in 2024. Assuming an α level of .05, what does the researcher conclude about the statistically significant difference?

t-statistic	df	p-value	95% CI Lower	95% CI Upper
-2.84	3616.79	0.005	-0.43	-0.08

(a) The results indicate a statistically significant increase in average daily TV viewing from the 1970s to 2024.
(b) Although there is a statistically significant difference, the direction of change is unclear.
(c) The confidence interval is below zero, indicating Americans watch less TV in 2024 than in the 1970s.
(d) The negative t-statistic suggests a decrease, meaning Americans watch statistically significantly less TV in 2024 than in the 1970s.

17. A sociologist finds that two neighbourhoods have the same average level of trust in local government. Residents in Neighbourhood A report trust scores ranging from 4 to 6, while residents in Neighbourhood B report scores ranging from 1 to 10. Which statistic would best summarize this difference?

(a) standard deviation
(b) median
(c) t-test
(d) mean

18. Researchers calculated a 95% confidence interval of the average family size of Canadians to be from 2.7 to 3.1, based on a sample of 500 families with a standard deviation of 1.1. What change would produce a more precise estimate?

(a) Increasing our confidence interval to 99%
(b) Increasing our sample size to 600
(c) Improving our sampling strategy
(d) Increasing our standard deviation to 1.3

19. Which of the following best describes the differences between the Console, R Scripts, and Quarto documents in RStudio?

(a) R Scripts and Quarto documents both run code line-by-line, but only the Console saves your work.
(b) The Console runs code interactively, R Scripts store reusable code, and Quarto combines code, text, and output for dynamic reports.
(c) The Console is used for writing long-term code, R Scripts are for temporary testing, and Quarto is only for plotting.
(d) The Console and R Scripts are identical, while Quarto is used only for Markdown formatting.

20. Which of the following code snippets correctly uses `mutate()` to convert height from centimeters to meters?

    (a) mutate(height <- height / 100)
    (b) mutate(height = select(height / 100))
    (c) mutate(height = height / 100)
    (d) mutate(height / 100 = height)

21. Which command installs a new package in R?

    (a) `install.packages()`
    (b) `require()`
    (c) `library()`
    (d) `load()`

22. A sociologist wants to examine whether there is a relationship between level of education (measured in years) and annual income (in dollars) among adults in the U.S. Which statistical test is most appropriate?

    (a) Pearson's correlation (r)
    (b) Logistic regression
    (c) t-test
    (d) Chi-square test

23. A researcher summarized the `sibs` variable from the U.S. General Social Survey, a measure of how many brothers and sisters respondents had. Based on the summary table, the distribution appears to be:

freq	median	mean	sd
3962	3	2.9	1.8

    (a) negatively skewed
    (b) positively skewed
    (c) nominal
    (d) symmetrical

24. A researcher compares support for universal healthcare (Support vs. Oppose) between two racial groups: Black and White respondents. Which statistical test should be used?

    (a) t-test
    (b) Chi-square test
    (c) Pearson's correlation (r)
    (d) Linear regression

25. Using the U.S. General Social Survey data, a researcher tests whether the proportions who favor a law requiring a person to obtain a police permit before they could buy a gun have changed over time. Assuming an α level of .05, which of these statements best summarizes the researcher's conclusion?

t-statistic	df	p-value	95% CI Lower	95% CI Upper
1.82	3114.93	0.069	-0.00	0.06

(a) There is a statistically significant difference in support for the police permit law between 1984 and 2024, but it was due to sampling error and cannot be interpreted.
(b) There is a statistically significant difference in support for the police permit law between 1984 and 2024, suggesting that public opinion has changed over time.
(c) There is a statistically significant increase in support for the police permit law from 1984 to 2024, proving that stricter gun control laws are now favored by most Americans.
(d) There is not a statistically significant difference in support for the police permit law between 1984 and 2024, indicating that public opinion has remained stable.

26. If a researcher concluded that there was no difference in police use of force against Black and White men when there actually is a difference, then:

(a) The null hypothesis was correctly retained.
(b) The null hypothesis was incorrectly retained (false negative).
(c) The null hypothesis was correctly rejected.
(d) The null hypothesis was incorrectly rejected (false positive).

27. A p-value is

(a) An estimate of the standard error of a sampling distribution
(b) A measure of how unusual or rare our obtained statistic is compared with what is stated in the null hypothesis
(c) An estimate of where our population statistic lies on the true sampling distribution of the means
(d) A measure that tells us the variability in a normal distribution

28. Which of these datasets has a standard deviation of 0?

(a) 2, 2, 5, 7, 7
(b) 1, 4, 7, 10, 15
(c) 4, 4, 4, 4, 4
(d) 0, 2, 4, 6, 8

29. Which statement best describes the result of this code?

```
df_couples <- left_join(df_partner1, df_family, by = "coupleid")
```

(a) Both datasets are appended by adding the columns of `df_family` to the end of `df_partner1`.
(b) It merges the datasets by matching on `coupleid`, but keeps only observations that appear in both datasets.
(c) The datasets are merged using `coupleid`, adding new columns from `df_family` to matching observations in `df_partner1`.
(d) The datasets are appended, adding the rows of `df_family` below the rows of `df_partner1`.

30. Based on the table, which statement best compares the relative frequency of being 'not too happy' between men and women?

sex	male	female	Total
happy			
very happy	350 (19.8%)	441 (20.5%)	791 (20.1%)
pretty happy	1025 (57.8%)	1264 (58.7%)	2289 (58.3%)
not too happy	397 (22.4%)	450 (20.9%)	847 (21.6%)
Total	1772 (100.0%)	2155 (100.0%)	3927 (100.0%)

(a) Women have a slightly higher relative frequency because more women than men reported being 'not too happy'.
(b) A greater number of women reported being 'not too happy' than men, so women have a higher relative frequency.
(c) Men have a slightly higher relative frequency of being 'not too happy' than women, even though fewer men reported it.
(d) The frequency of being 'not too happy' is equal for men and women.

31. Which of the following statements best reflects the ASA's guidance on interpreting p-values?

(a) p-values measure the scientific or economic significance of a result.
(b) p-values are influenced by sample size and measurement precision, and do not reflect effect size or importance.
(c) A large p-value proves that no effect exists.
(d) A small p-value always indicates a large and important effect.

32. What does the following R code produce?

```
gss21 |>
  tbl_cross(
    row = incgap,
    col = sex,
    percent = "column",
    missing = "no") |>
  add_p()
```

(a) A frequency table of `incgap` responses with missing values imputed and grouped by `sex`
(b) A summary table of `sex` grouped by `incgap`, with missing values excluded and a t-test.
(c) A cross-tabulation of `incgap` by `sex`, showing column percentages and a chi-square test of independence
(d) A cross-tabulation comparing the mean values of `incgap` between men and women using a t-test

33. Why is it considered good practice to use RStudio Projects when working on data analysis?
    (a) They ensure your code runs faster by optimizing memory usage
    (b) They prevent syntax errors by enforcing tidyverse formatting rules
    (c) They help organize your files, set a working directory, and keep your environment consistent
    (d) They automatically load packages when you open RStudio

34. What does the following code return?

```
gss24 |>
  group_by(sex) |>
  summarize(mean_age = mean(age, na.rm = TRUE))
```

    (a) A new column named `mean_age` added to the original data frame.
    (b) A data frame showing the average age for each sex.
    (c) A filtered data frame with only rows where age is not missing.
    (d) A list of all ages grouped by sex.

35. According to the 68-95-99.7 rule, what can we say about a person whose score is more than 2 standard deviations above the mean?
    (a) Their score is within the middle 68% of the distribution.
    (b) Their score is likely below average.
    (c) Their score is very common.
    (d) Their score is unusual and higher than most.

36. Using 2024 GSS data, a researcher estimated how many Americans believe in life after death. Based on the 90% confidence interval from R, which interpretation is correct?

n	mean	sd	lowCI	hiCI
1283	0.83	0.38	0.81	0.85

    (a) The researcher is 90% confident that the proportion of U.S. adults who believe in some form of life after death is between .81 and .85.
    (b) If another survey had been conducted in 2024, there's a 90% chance the point estimate would be between .81 and .85.
    (c) Because the mean is within the confidence interval, there is a .90 probability that the proportion of U.S. adults who believe in some form of life after death is .83.
    (d) About 90% of people responded to the national survey.

37. If the final grades in a statistics class were normally distributed with an average of 81.5% and a standard deviation of 1.5, about what proportion of the students' final grades were between 80% and 83%?
    (a) 50%
    (b) 62%
    (c) 95%
    (d) 68%

38. Using the table, which of the following is the correct formula to find the expected frequency for U.S. liberals who oppose the death penalty in 2024?

ideology	conservative	liberal	Total
cappun			
favor	573 (63.0%)	337 (37.0%)	910 (100.0%)
oppose	204 (32.4%)	425 (67.6%)	629 (100.0%)
Total	777 (50.5%)	762 (49.5%)	1539 (100.0%)

(a) (425 * 762)/1539
(b) (425 * 1539)/629
(c) (629 * 762)/1539
(d) (629 * 425)/1539

39. What will be true of `gss24` after this code is run?

```
gss24 |>
  filter(age > 20) |>
  select(age, childs)
```

(a) `gss24` will contain only rows with `age` greater than 20 and columns `age` and `childs`.
(b) `gss24` will contain only respondents older than 20 and who are parents.
(c) `gss24` will be unchanged.
(d) `gss24` will contain only the variables `age` and `childs`.

40. What is Wheelan's primary message of Chapter 1 ("What's the Point?") in *Naked Statistics: Stripping the Dread from the Data?*

(a) Statistics explain why relationship exist between variables
(b) Statistical formulas are the most important part of understanding data
(c) Statistics provide honest answers to complex questions
(d) Statistics simplify complex information to help us make better decisions

41. According to Cohen's argument in *Citizen Scholar*, what role should descriptive statistics play in social science research?

(a) Descriptive work should be minimized in favor of studies that make strong theoretical contributions and advance broader explanatory frameworks.
(b) Descriptive statistics are a foundational part of scientific inquiry and should be valued as legitimate research.
(c) Descriptive statistics are predominantly useful as a step before hypothesis testing, rather than as a meaningful research contribution on their own.
(d) Descriptive statistics can summarize patterns in data, but rigorous social science research should focus primarily on inferential methods.

42. Which function is used to load a package in R?

    (a) `library()`
    (b) `source()`
    (c) `install.packages()`
    (d) `load()`

43. What will the following code do?

```
gss24 <- gss24 |>
  mutate(status = case_when(
  income >= 100000 ~ "High",
  income > 50000 & income < 100000 ~ "Medium",
  TRUE ~ "Low"
))
```

    (a) It creates a new column named `status` with 3 possible values.
    (b) It filters out rows where income is less than 50,000.
    (c) It replaces the numeric values in the income column with text labels.
    (d) It removes missing values from the income column.

44. What will happen if you try to use a function from a package that hasn't been loaded?

    (a) R will prompt you to install the package
    (b) The function will run normally
    (c) You will get an error saying the function is not found
    (d) R will automatically load the package

45. What does the pipe operator (|>) do in dplyr?

    (a) It creates a new column in a data frame.
    (b) It passes the result of one function into the next.
    (c) It filters rows based on a condition.
    (d) It loads the dplyr package into memory.

46. A respondent reports 5 poor mental health days in the past 30 days. Based on summary stats for the `mntlhlth` variable, which of the following best describes their z-score?

freq	median	mean	sd
2279	0	4.5	7.9

    (a) The z-score is large and positive because 5 is much higher than the mean.
    (b) The z-score is undefined because the median is 0.
    (c) The z-score is close to zero because 5 is close to the mean.
    (d) The z-score is negative because 5 is below the sd.

47. Chi-square values cannot be _____.

    (a) negative
    (b) positive
    (c) zero
    (d) outside the range of 0 to 1

48. Based on the table, which description best describes the correlation between personal income (`realrinc`) and number of days of poor mental health (`mntlhlth`)?

Correlation Coefficient	t-statistic	df	p-value	95% CI Lower	95% CI Upper
-0.114	-5.04	1933	0.000	-0.16	-0.07

    (a) Decreases in personal income was correlated with decreases in the number of days of poor mental health ($t = -5$, $p < .05$).
    (b) Results indicate an inverse relationship ($r = -0.11$, $p < .05$), suggesting that the number of days of poor mental health decreases as personal income increases.
    (c) There is no evidence of an association between personal income and days of poor mental health ($t = -5$).
    (d) There is a weakly positive association between personal income and days of poor mental health ($r = -0.11$, $p < .05$).

49. Wheelan illustrates what key concept in the chapter "Inference: Why My Statistics Professor Thought I Might Have Cheated?" from the book *Naked Statistics: Stripping the Dread from the Data*?

    (a) That unusually accurate results can raise suspicion, but statistical inference helps assess whether they are likely due to chance
    (b) That statistical inference can be used to prove someone's guilt beyond reasonable doubt
    (c) That cheating can be detected by comparing mean scores across different groups
    (d) That hypothesis testing is not useful in criminal investigations

50. In *White Logic, White Methods*, what does Gandy suggest about the use of racial statistics?

    (a) Their use in legal proceedings increasingly lead to unfair policy outcomes
    (b) They are neutral tools most useful for academic research
    (c) Their use in data journalism helps mitigate online polarization
    (d) They can both expose and perpetuate discrimination